



Prepared with Synoptic

Preliminary investigation report

Grounding on Boolarong Bank, Merrandi Channel, Kanooka Heads

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Summary

On 07/07/2026 the passenger ferry MV Tarrakin Star left the dredged Merrandi Channel at Kanooka Heads and grounded on Boolarong Bank at 18:12:04. It was inbound on service TF18 in darkness with 38 passengers embarked. There were no serious injuries, although one passenger reported a grazed shin and two were treated at the wharf for distress. The starboard hull bottom was split, the starboard propeller, shaft, rudder stock and skeg were damaged, the vessel was withdrawn from service, and the channel was closed to commercial traffic for 19 hours. The investigation is examining the chart plotter freeze before the vessel left the channel, how night operation of the channel was assessed and planned, the visibility of the No. 4 beacon and the shallow water alarm setting. In the meantime, the mangrove canopy screening the beacon has been trimmed and a local notice giving the beacon's changed range has been issued.

What we know

On 07/07/2026 the MV Tarrakin Star, a 24.4 m aluminium catamaran passenger ferry certificated for 96 passengers, was operating Tarrakin Ferries service TF18 from Muntari Island to Kanooka Heads. TF18 was the last service of the day. Since 01/06/2026 the winter timetable has placed it in darkness. On board were 38 passengers, a master, a deckhand and a kiosk attendant.

The masters changed over at the town wharf at about 13:05. The outgoing master told the relieving master that the chart plotter display had frozen twice that morning and had recovered by itself each time. The plotter's event log records those freezes at 10:42 and 11:17. Neither master entered them in the defect book.

TF18 left Muntari Island at 17:35, after the end of civil twilight at 17:29. The vessel entered the Merrandi Channel from the north at 18:03:58. By 18:05:30 it was on the charted centreline abeam the No. 2 beacon, at 8.4 kt. The deckhand had left the bridge at 18:04 to collect fares.

At 18:07:06 the plotter display froze again, and this time the log records no recovery. Over about the next minute the recorded heading increased from 195 to 200 degrees and was then held. Engines, steering hydraulics and rudder response stayed within normal limits throughout. At 18:08:31 the master began looking for the No. 4 beacon. At about 18:09 the vessel crossed the western edge of the dredged channel. The master sighted the beacon at 18:10:02, when the vessel was 39 m west of the centreline. The course then eased about three degrees back to port.

The echo sounder's shallow alarm, set at 0.9 m under the keel, sounded at 18:11:48. Both throttles went to astern four seconds later. At 18:12:04 the vessel struck Boolarong Bank at 6.8 kt, 74 m west of the centreline and 54 m outside the dredged channel. It stopped within 11 m on a falling tide.

The master stopped both engines at 18:12:31 and did not attempt to refloat. At 18:13:04 he broadcast a pan pan to Kanooka Heads Port Control on VHF channel 16 and asked for the port workboat. All passengers were in lifejackets by about 18:22. The workboat came alongside at 18:41, and the last passenger was ashore at 19:26.

There were no serious injuries. One passenger reported a grazed shin, and two were treated at the wharf for distress. The starboard hull bottom plating was set up over 4.1 m with a 180 mm split. Water in the No. 2 void rose from approximately 75 mm at 18:16 to approximately 260 mm at refloating. The starboard propeller and shaft were bent, the rudder stock was distorted and the skeg was torn away. The port side was undamaged. The vessel refloated at 00:20 on 08/07/2026 and was withdrawn from service. The channel was closed to commercial traffic for 19 hours.

What we are looking at

- 1. The chart plotter freeze.** The investigation is examining why the plotter display froze at 18:07:06 and did not recover. This includes a manufacturer firmware update associated with the fault code the unit logged, which had never been applied to this unit. It is also examining how earlier freezes that cleared themselves were handled under the operator's defect procedure, OPS-9, and at the change of master.
- 2. Night operation of the channel.** The investigation is examining how the winter timetable change, CHG-2026-014, was assessed against the channel's route risk assessment, RRA-07, which restricts transits to daylight. It is also examining why the passage plan gives no guidance for losing the plotter and names no hold point (a set point at which to stop). A formal interview with the Operations Manager on that assessment has not yet been taken.
- 3. The No. 4 beacon.** The investigation is examining how visible the beacon was from the channel at night. This includes mangrove growth on Merrandi Islet, the repeated deferral of work order WO-118 to clear the sightline, and the beacon's solar conversion in 11/2025, of which channel users were not told. The effect on the night of new boat ramp floodlights behind the beacon has been examined and could not be resolved on the evidence available.
- 4. Actions on the bridge and the duty behind them.** The investigation is examining the heading change after the freeze and the continuation of the transit at speed. It is also examining the practice of the deckhand leaving the bridge to collect fares, and a winter roster that places the darkness transit at the end of a split duty. Whether the master was fatigued has been examined and could not be resolved.
- 5. The shallow water alarm.** The investigation is examining the echo sounder's alarm setting of 0.9 m under the keel, unchanged since 2019, against the vessel's stopping distance and the channel's depths.
- 6. Lifejacket access.** The investigation is examining a kiosk counter fitted in 2019 that blocks access to the forward saloon lifejacket locker. Questions about the operator's two other 24 m vessels are with the operator and the state regulator.

Containment and next steps

Interim action and containment

- The General Manager Operations suspended the evening Muntari services on 08/07/2026, and the vessel was withdrawn from service after the grounding. [NOT ESTABLISHED: whether the suspension of evening services remains in place.]
- The Port Authority funded work order WO-176 from the 2026 waterways contingency. Under it, the Merrandi Islet mangrove canopy was trimmed to 2 m on 24/07/2026, which restored the No. 4 beacon's sightline over the northern approach. A three-yearly trimming cycle has also been programmed. This action is complete.
- Local notice to waterway users LNM-2026-14 was issued on 22/07/2026, giving the No. 4 beacon's solar conversion and its 3 M nominal range. In addition, the aids to navigation register has been made available to every commercial operator using the channel. Tarrakin Ferries, for its part, has reissued its passage plan appendix as issue 3 with the corrected range. This action is still being implemented.
- The investigation has asked Tarrakin Ferries to write a night passage plan for the channel before any further darkness transit. That plan would name a hold point if the primary means of position fixing is lost. Separately, the investigation has asked that the services agreement replace its 50 m vegetation scope with a sightline test. Responses to both requests were due by 30/09/2026, and both remain open.
- No interim action has been recorded on the forward saloon lifejacket locker.

Next steps

Several pieces of work remain for the investigation. It still has to:

- take a formal interview with the Operations Manager on how CHG-2026-014 was assessed
- examine what caused the plotter freeze, which a 46 hour bench test could not reproduce
- obtain the operator's and the state regulator's answers on the two other 24 m vessels
- give the people and organisations concerned the chance to correct matters of fact, and have the draft read by someone outside the investigation.

The boat ramp floodlights question also remains open. [NOT ESTABLISHED: expected date of the final report.]

Distribution

Title	Name
General Manager Operations	H. Vardon
Safety Manager	P. Nandasiri
Operations Manager	C. Bickerton
Harbour Master, Kanooka Heads	L. Truscott
Aids to Navigation Officer, Kanooka Shire Council	E. Maguire